# Mahmoud Alyosify

NLP Engineer | Language Model Fine-Tuning, Text Understanding & Evaluation
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## PROFESSIONAL SUMMARY

NLP Engineer working across the full language stack — **tokenisation and decoding, domain fine-tuning, information extraction, semantic search and evaluation**. Currently an **AI Engineer Intern at Injaz Tech**, building retrieval-augmented question answering with hybrid search, cross-encoder re-ranking and query rewriting. Master's research at **Queen's University, Canada** (expected **Nov 2026**, **Digilians Presidential Scholarship**) builds a training-free decoding method evaluated on **summarisation (XSum)**, **extractive QA (SQuAD-v2)** and **open-ended generation (MT-Bench)** using **BERTScore, ROUGE, exact-match/F1** and a length-controlled LLM judge. Fine-tuned **Meta-Llama-3-8B** on a **159,826-record** domain corpus, and built an **entity-and-relation extraction pipeline** that converts unstructured open-source text into a queryable knowledge graph. Grounded in the part that decides whether NLP ships: evaluation that does not flatter the model.

## EDUCATION

**MSc in Artificial Intelligence** — *Sep 2025 – Nov 2026*
*Queen's University, Kingston, Ontario, Canada* — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

**B.Sc. Computer & Information Science (Bioinformatics)** — *Sep 2019 – Jul 2023*
*Assiut University, Egypt* — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

## TECHNICAL SKILLS

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|---|---|
| **NLP Core** | Transformers, Tokenisation, Text Summarisation, Extractive Question Answering, Information & Relation Extraction, Knowledge-Graph Construction, Text Classification, Semantic Similarity & Redundancy Detection, Sentence Embeddings, Semantic Search, RAG, Hybrid Search (BM25 + Dense, RRF), Cross-Encoder Re-Ranking, Query Rewriting |
| **Language Models** | Instruction Fine-Tuning, LoRA, QLoRA, PEFT, Custom Decoding Strategies, Prompt Engineering, Structured Outputs, Self-Hosted Inference · *Mistral-7B, Llama-3.1-8B, Meta-Llama-3-8B, Qwen2.5-3B* |
| **Frameworks** | PyTorch, Hugging Face Transformers / Datasets / PEFT, TensorFlow, Scikit-learn, NLTK, LangChain, LangGraph, sentence-transformers, FAISS, ChromaDB, Pandas, NumPy |
| **Evaluation** | BERTScore, ROUGE, Exact-Match / F1, LLM-as-Judge, MT-Bench, XSum, SQuAD-v2, Non-Inferiority Testing, Ablation Studies, Reproducible Benchmarking |
| **Deployment** | AWS, Docker, FastAPI, Streamlit, ONNX, CI/CD, Git/GitHub, Linux, SQL |
| **Programming** | Python, SQL, C++, C#, JavaScript, R, Bash |

## PROFESSIONAL EXPERIENCE

**AI Engineer Intern** — *Oct 2026 – Present*
*Injaz Tech — Injaz Group (formerly Fingerprint Consultancy)* — Maadi, Cairo, Egypt

- Built retrieval-augmented question answering over internal documentation with **LangChain** and **LangGraph**: header-aware Markdown chunking, sentence embeddings in **FAISS**, BM25 and hybrid RRF search, **cross-encoder re-ranking**, query rewriting and follow-up question contextualisation for multi-turn chat.
- Benchmarked retrieval with custom metrics (top-1 accuracy, MRR), raising top-1 accuracy from **75% to 94%**, and built an evaluation suite (LLM-as-judge correctness and groundedness plus deterministic key-fact checks) that runs as a CI regression gate.

**Machine Learning Instructor** — *Aug 2026 – Sep 2026*
*National Telecommunication Institute (NTI)* — Arish & Assiut, Egypt

- Rehired to teach two **90-hour** applied ML bootcamps (**50 trainees**) at the Arish (online) and Assiut branches — regression, SVM, KNN, Naive Bayes, PCA and neural networks with Python labs, final exams and capstone-project evaluation.

**Machine Learning Instructor** — *Jul 2025 – Sep 2025*
*National Telecommunication Institute (NTI)* — Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks with practical labs.

**AI in Cybersecurity Intern** *Jun 2025 – Jul 2025*
*e& Egypt — via NTI (On-the-Job Training)* Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

**Systems Engineer (Military Service)** *Feb 2024 – Mar 2025*
*National Company for Roads Building & Development* Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

**Online Instructor — AI, Algorithms & Data Science** *2020 – Present*
*Udemy & YouTube* Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

## SELECTED NLP PROJECTS

**Verbosity-Aware Decoding for Token-Efficient Language Model Inference** *2026 – Present*
*MSc Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi*

- Built a **training-free LogitsProcessor** operating directly on the decoding loop, combining **stop-token**, **n-gram-repetition** and **semantic-redundancy** signals under an **entropy-driven gate** to cut generated tokens without retraining or architecture change.
- Evaluated across three distinct NLP task types — **open-ended dialogue (MT-Bench, YapBench)**, **abstractive summarisation (XSum)** and **extractive QA (SQuAD-v2)** — on Mistral-7B, Llama-3.1-8B and Qwen2.5-3B, cutting generated tokens by **25.5–34.9%** on MT-Bench.
- Designed a **pre-registered protocol** proving semantic preservation through paired **non-inferiority testing** using **BERTScore, ROUGE, exact-match/F1** and a **length-controlled LLM judge** — controlling for the length bias that quietly inflates most generation metrics.

**Horus-OSINT — Domain-Adapted Language Model for Event Intelligence** *2026*
*Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python*

- Curated and formatted **159,826** GTD/GDELT records into an instruction dataset, then fine-tuned **Meta-Llama-3-8B** for natural-language querying of global event data.
- Deployed the adapted model on **AWS** as an interactive conversational assistant over structured and unstructured intelligence sources.

**HORUS Sentinel — Text-to-Knowledge-Graph Intelligence Pipeline** *2026*
*Information Extraction · Knowledge Graphs · Multi-Agent Systems · Fine-Tuned LLM*

- Built a pipeline that turns continuously collected **unstructured open-source text** into a **living knowledge graph**, correlating entities and relations across sources rather than treating each document in isolation.
- A self-hosted fine-tuned model reasons over the graph to generate **prioritised, evidence-backed reports** with every claim traceable to source text.

**Automated Multimodal Agent — Document Understanding to Narrated Output** *2026*
*Document Parsing · Text Generation · Speech Synthesis*

- Built an autonomous pipeline that extracts and restructures document content into slide-level summaries, then renders them as narrated audio — chaining extraction, generation and synthesis end to end.

### Additional Projects

- **SimCLR-Vision-SSL** — contrastive representation learning with **FAISS** embedding retrieval — the same vector-search machinery used in semantic search and RAG.
- **Human vs. Agentic Pull Requests** — large-scale text and metadata mining of GitHub pull-request discussions.
- **Also:** AudioShield (audio classification) · SCRAP · RL-Cloud-Autoscaler · Vitalism Solution (rPPG).

## CERTIFICATIONS & AWARDS

**AWS Academy** — Machine Learning for NLP
**Generative AI & Prompt Engineering** — IBM / Coursera
**HCIP – AI** (Huawei)
**1st Place** — Smart Cities Hackathon 2022
**AWS Academy** — ML Foundations, Data Engineering
**Machine Learning Specialization** — DeepLearning.AI
**Digilians Presidential Scholarship** — Queen's University
**Semi-Finalist** — Microsoft Imagine Cup 2023

## LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.